Derek Nagel

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PROFILE

Mechanical Engineering student with a concentration in Aerospace, currently a Mechanical Engineering Intern at Astronics Connectivity Systems & Certification working on aerospace test engineering, structural analysis, and manufacturing automation. Hands-on builder across mechanical, electrical, and software projects. Licensed *pilot* for single and multi-engine aircraft. Division I volleyball athlete, recognized for leadership and discipline in high-stakes environments.

EDUCATION

George Mason University — Fairfax, VA
B.S. Mechanical Engineering with Concentration in Aerospace — Expected Graduation — May 2027
Minor — Aviation Flight Training and Management — Completed

Relevant Coursework: Practicum in Engineering, Dynamics, Differential Equations, Statics, Thermodynamics, Solid Mechanics, Air Transportation System Engineering, Human Factors

Honors: Presidential Academic Scholarship

TECHNICAL SKILLS

SolidWorks, MATLAB, Autodesk Inventor, Python, Excel & Data Visualization, AI-Assisted Engineering Automation (Claude), EV & HV Electrical Systems, Private Pilot (Single & Multi-Engine)

PROFESSIONAL EXPERIENCE

Astronics Connectivity Systems & Certification — *Mechanical Engineering Intern* — May 2026 – Present
Waukegan, IL

- Support **aerospace LRU mechanical design** for in-flight connectivity and entertainment hardware flying on Airbus and Boeing platforms, within AS9100 quality processes.
- Converted a 4-MCU ARINC 600 tray assembly into a SolidWorks structural model for **modal analysis** to DO-160G qualification requirements.
- Led recovery of a Qualmark Typhoon 3.0 **HALT environmental test chamber** from an extended outage; root-caused the fault and returned accelerated-life testing to service.
- Built the **capital justification for a proposed robotic riveting cell** — process time studies, throughput modeling, and CAPEX/payback analysis — presented to engineering leadership and informed by vendor research at AUTOMATE 2026.

Tesla — *Tesla Advisor (Seasonal)* — May 2025 – Sep 2025
Schaumburg, IL

- Supported **end-to-end electric vehicle delivery operations**, including late-stage vehicle staging, quality checks, and final inspection to ensure safety, performance, and delivery readiness.
- Explained **technical EV systems** (battery technology, drivetrain, charging, and software features) to customers, translating complex engineering concepts into clear, user-focused communication.
- Shadowed operations at Tesla's **Elgin manufacturing facility**, observing battery production, electric motor components, and assembly-line processes, quality control, and process sequencing.

Independent Study *in Electrical and Mechanical Design* — Aug 2022 – Present

- *Tesla Model S Re-build* — Disassembled and rebuilt a heavily corroded battery pack and drivetrain on a 2014 Tesla Model S P85 (260,000 miles); refurbished the pack with a custom corrosion treatment and returned the car to the road.
- *Plane Pointer* — Designed and built a 2-axis automated aircraft tracking system that physically points toward live aircraft using ADS-B data while displaying flight ID, position, altitude, and speed.
- *Pre-AP Steering CAN Toolkit* — Reverse-engineered the CAN protocol of a Tesla electric power-steering rack and built a Python bench-control toolkit toward retrofitting driver assistance onto pre-Autopilot vehicles.
- *3D LED Display System* — Created a dynamic volumetric light display using custom circuits and optical principles; matrix, control circuitry, and patterns built from scratch.

EXTRACURRICULAR

Division I Athlete, Men's Volleyball — George Mason University — compete in a demanding D1 program while carrying a full engineering course load.

President, Athletes in Action — George Mason University — lead a student-athlete ministry chapter: weekly gatherings, service projects, and mental-health outreach across teams.